

PRANAV KHERA

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SUMMARY

Data Science & Machine Learning–focused dual-degree undergraduate (B.Tech Information Technology + B.Sc. Data Science and AI, IIT Guwahati) with hands-on experience building end-to-end ML systems — spanning graph-based fraud detection, algorithmic trading models, and on-device mobile ML. Proficient in Python, XGBoost, Scikit-learn, and applied ML pipelines, with proven ability to integrate LLM APIs (Gemini, Claude) into production-style systems. Strong foundation in Data Structures & Algorithms and full-stack development (React/Next.js, REST APIs). Seeking a Data Science/Machine Learning or Software Engineering internship.

EDUCATION

Indian Institute of Technology (IIT) Guwahati — B.Sc. (Hons.) Data Science & Artificial Intelligence CGPA: 8.50/10 (Semester 2) 7.50/10 (Semester 1)	Expected 2029
Guru Tegh Bahadur Institute of Technology (GTBIT) — B.Tech Information Technology CGPA: 8.76/10	Expected 2029
Modern Vidhya Niketan School, Sec-17 — Class XII (CBSE)	Completed

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C++, C, SQL, Kotlin

Machine Learning: XGBoost, Regression, Classification, Clustering, Isolation Forest, Feature Engineering, Explainable AI (SHAP), On-Device ML (TensorFlow Lite)

Data Science: Pandas, NumPy, Scikit-learn, Exploratory Data Analysis (EDA), Data Visualization

Web & Mobile: React, Next.js, FastAPI, Flask, Tailwind CSS, Android (Kotlin, Jetpack Compose), React Three Fiber

Databases & Backend: Firebase, Firestore, Supabase, PostgreSQL, NetworkX

AI / LLM Integration: Gemini API, Claude API

Tools & Practices: Git/GitHub, Data Structures & Algorithms (LeetCode), Version Control

KEY PROJECTS

TradingBOT — Algorithmic Paper-Trading Engine Python, XGBoost, Flask, yfinance

- Built a real-time algorithmic paper-trading system for Indian equities (RELIANCE, TCS, INFY, HDFCBANK) with a 2-second live scalping scheduler and a simulated broker (trailing stop-loss, smart averaging-down).
- Engineered an 18-feature XGBoost signal model using technical indicators (RSI, MACD, Bollinger Bands, ATR) trained on historical intraday candles, combined with a live news-sentiment filter to adjust trade confidence.
- Built a Flask-based live dashboard visualizing equity curves, active positions, and closed-trade ledgers in real time.

InvoiceIQ — AI Financial Intelligence Platform Next.js, TypeScript, XGBoost, Gemini API, Claude API, Supabase

- Designed a fintech platform automating invoice parsing, GST compliance, and reconciliation for Indian SMEs, using the Gemini and Claude APIs for document understanding and verification.
- Built a 4-layer fraud-detection pipeline (rule-based → XGBoost ensemble → AI verification) reaching 96%+ accuracy and sub-150ms inference on transaction data.
- Implemented multi-currency support, invoice-financing workflows, and a Supabase/PostgreSQL backend.

ScamShield — On-Device Android Scam-Call Detector Kotlin, Jetpack Compose, TensorFlow Lite

- Built a privacy-first Android app that analyzes live phone calls using on-device speech-to-text and a TFLite ML model layered with rule-based pattern matching across 10 scam categories.
- Designed the system to surface real-time risk alerts mid-call and discard all captured audio/text immediately after the call ends (zero data retention).
- Motivated by firsthand encounters with fraudulent internship and job-offer schemes; aimed at protecting users from real-time voice scams.

Fraud-Detector Engine Python (FastAPI, NetworkX, SHAP), Next.js, Tailwind CSS, Gemini API

- Engineered a real-time, graph-based fraud detection engine applying anomaly detection and behavioral data analysis to flag suspicious activity.
- Integrated the Gemini API to generate natural-language explanations for flagged fraud cases, improving interpretability of model outputs.
- Built a responsive full-stack interface using Next.js and Tailwind CSS to visualize explainable anomaly-detection results.

CivicPulse — Hyperlocal Civic Issue Reporting Platform Next.js, TypeScript, Gemini Vision API, Leaflet

- Built a full-stack civic-engagement platform where citizens report infrastructure issues (potholes, sanitation, safety hazards) with photo/video evidence and geolocation.
- Integrated Gemini Vision to auto-classify issue type and severity from images, and to verify before/after photos when volunteers mark issues resolved.
- Designed a gamified engagement system (XP, badges, leaderboards) and a policy-advisory console summarizing neighborhood reports into municipal recommendations.

Other Projects (GitHub): Stellaris — interactive 3D star map & constellation visualizer (Next.js, React Three Fiber); Payout-App — milestone-based escrow platform for startup–investor fintech (Next.js, Firebase, Flask); Macros Analyzer — AI nutrition tracker using the Gemini API (Python, Firebase).

CERTIFICATIONS

- Fundamentals of Python Programming — E&ICT Academy, IIT Kanpur (MeitY initiative), Oct 2025
- BSc (Hons.) Data Science & AI — Orientation Course, IIT Guwahati (via Coursera), May 2026

ACHIEVEMENTS, EXTRACURRICULARS & COMPETITIVE PROGRAMMING

- Actively solving DSA problems on LeetCode (35+ solved — Trees, Graphs, DP, Two Pointers); tracked at github.com/Pranav2-4-7/LeetCode
 - CTF cybersecurity participant (DTU); active member, IEEE GTBIT Student Branch (IEEE Day 2025, TARANG 2.0, HackTivate)
 - Hackathons: Vibe2Ship (Coding Ninjas × Google for Developers), AI Solution Expo (Bharati Vidyapeeth's College of Engineering)
- Languages: English (Professional), Hindi (Native), Punjabi (Native)